

Customer Shopping Behavior Analysis

Turning 3,900 retail transactions into strategic insight with Python, SQL, and Power BI.

DATA ANALYTICS PORTFOLIO PROJECT

CLASS PRESENTATION · END-TO-END ANALYSIS

Find the patterns that tell a retailer where to spend next

SPENDING PATTERNS · CUSTOMER SEGMENTS · PRODUCT PREFERENCES · SUBSCRIPTION
BEHAVIOR.

3,900 purchases · 18
columns

PYTHON · SQL · POWER BI

Today's journey through the analysis

1 The dataset

What 3,900 transactions and 18 columns actually contain.

2 The approach

Cleaning in Python, then business queries in PostgreSQL.

3 Key findings

Ten business questions grouped into four themes.

4 The standout insight

Where loyalty and subscription reinforce each other.

5 The dashboard

Power BI views built for revenue, segments, and products.

6 Recommendations

Where the business should act first, and why.

The dataset is small enough to trust and rich enough to segment

TRANSACTIONS

3,900

Individual purchases analyzed

COLUMNS

18

Features before engineering

MISSING RATINGS

37

Imputed by category median

SIGNAL GROUPS

3

Demographics · purchase · behavior

Demographics, purchase details, and shopping behavior sit in one table — so every revenue question can be cut by customer at the same time.

02

From raw data to business answers

The analysis moved in one direction: clean the file in Python, ask the business questions in SQL, then make the answers visible in Power BI.

[CLEANING](#) · [SQL ANALYSIS](#) · [DASHBOARD](#)

Four preparation steps made the data answerable

1 Clean in Python with pandas

Loaded and explored the file, imputed 37 missing review ratings by category median, standardized every column to snake_case.

2 Engineer the features that segment customers

Added age_group bands and purchase_frequency_days so demographic and cadence questions could be asked directly in SQL.

3 Check consistency and cut redundancy

Dropped promo_code_used once it proved redundant against discount_applied, removing a duplicate signal from the discount analysis.

4 Load into PostgreSQL

Moved the cleaned table into a database so the ten business questions ran as real, repeatable SQL rather than one-off scripts.

Ten SQL questions, grouped into four decisions

LENS ↓ FOCUS →	WHO SPENDS	WHAT THEY BUY
REVENUE	 Revenue drivers Revenue by gender and by age group — where the money actually comes from.	 Product performance Top 5 products by rating, top 3 per category, and the items that only sell on discount.
CUSTOMER	 Customer value High-spending discount users, and subscriber versus non-subscriber spend.	 Loyalty & segmentation New, Returning, and Loyal segments, repeat-buyer subscription likelihood, and standard versus express shipping.

Loyalty and subscription reinforce each other.

Customers with five or more previous purchases are markedly more likely to hold a subscription — and subscribers show both higher average spend and higher total revenue than non-subscribers. Repeat buying and subscribing are the same behavior seen from two directions, which makes either one a lever on the other.

Visualize the findings, then act on three fronts



Visualize

- Power BI dashboard surfacing revenue, segments, and product performance.
- One interactive view per business question.

OUTPUT · DASHBOARD



Act

- Boost subscriptions with exclusive member benefits.
- Launch loyalty rewards that move repeat buyers into the Loyal segment.
- Rebalance discount policy to protect margin.

OUTPUT · RETENTION PLAN



Target

- Spotlight top-rated and best-selling products in campaigns.
- Focus spend on high-revenue age groups and express-shipping customers.

OUTPUT · CAMPAIGN FOCUS

Grow subscriptions and reward loyalty first.

BACKED BY

- 1 Disciplined discount policy
- 2 Top-rated products in campaigns
- 3 Targeted marketing by age group

TAKEAWAY

Subscription growth and loyalty programs are the highest-leverage next moves.